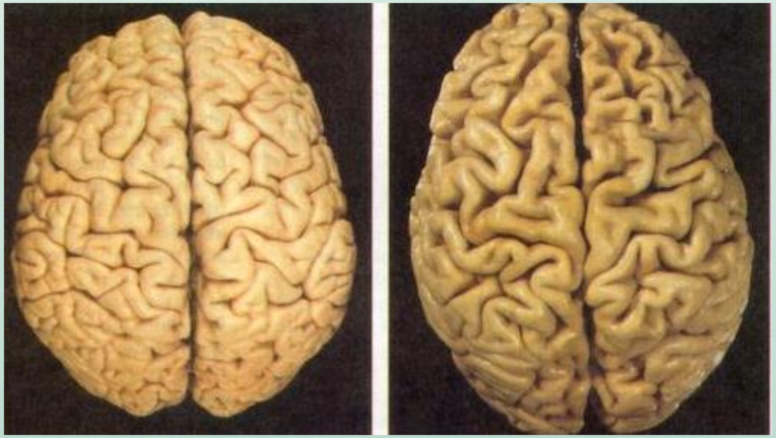
APPROACH TO DEMENTIA



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Outline:

- Definition
- Dementia vs Delirium
- Subtypes of Dementia
- Cortical vs Subcortical Dementia
- Mixed Dementia
- Workup of Dementia

Dementia:

- An acquired syndrome of intellectual impairment that interferes with social & occupational function
- Associated with impairment in at least one or more of the following cognitive domains:
  - ✓ learning and memory,
  - ✓ language,
  - ✓ executive function,
  - ✓ complex attention,
  - ✓ perceptual-motor, or
  - ✓ social cognition

Mild cognitive impairment (MCI):

- **MCI** is an intermediate clinical state between normal cognition and dementia.
- While specific subtle changes in cognition can occur in normal aging, MCI can also be a **precursor** to dementia.
- MCI may also represent a **reversible** condition in the setting of depression, as a complication of certain medications, or during the recovery from an acute illness.

Dementia Vs Delirium

- **Delirium**: characterized by a disturbance of consciousness and a change in cognition that develop over a short period of time

- Delirium will develop in about:
  - 10-15% of surgical inpatients.
  - **15-25**% of medical inpatients.
  - 30% of ICU patients.

Dementia Vs Delirium

Delirium	Dementia
Acute	**Chronic insidious**
Registration problem	**Recall problem**
Wax &Wane	**No fluctuation**
Hallucination common	Not common
Agitation present	Not present
Systemic manifestations present	Not found

Reversible **DEMENTIA**:

D= Drugs, Delirium, Depression (3 Ds)

E= Endocrine Disorders*

M= Metabolic Disturbances*

E= Eye & Ear Impairments

N= Nutritional Disorders

T= Tumors, Toxicity*, Trauma to Head (3 Ts)

I= Infectious Disorders

A= Alcohol, Arteriosclerosis

Irreversible DEMENTIA:

- Alzheimer’s
- Lewy Body Dementia
- Frontotemporal Dementia (Pick’s Disease)
- Parkinson’s disease (& Other Parkinsonism's)
- Heady Injury
- Huntington’s Disease
- Cruzefeldt-Jacob Disease

Anatomical substrates of memory in Dementia?

Medial temporal lobes & hippocampus (also fornix, mamillary bodies of limbic system)

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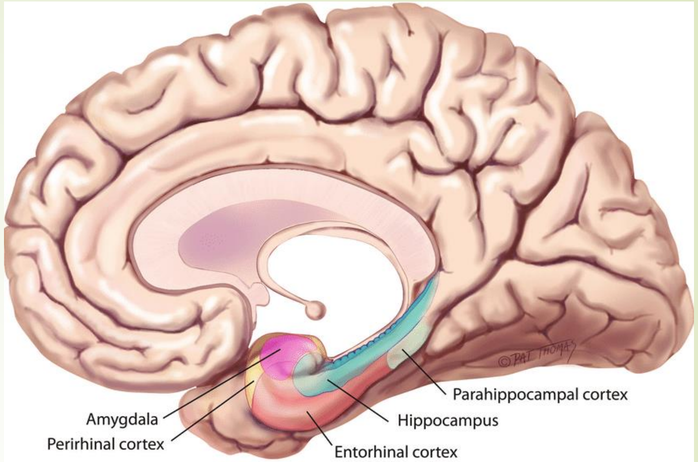

Medial thalamus - especially dorsal median nucleus

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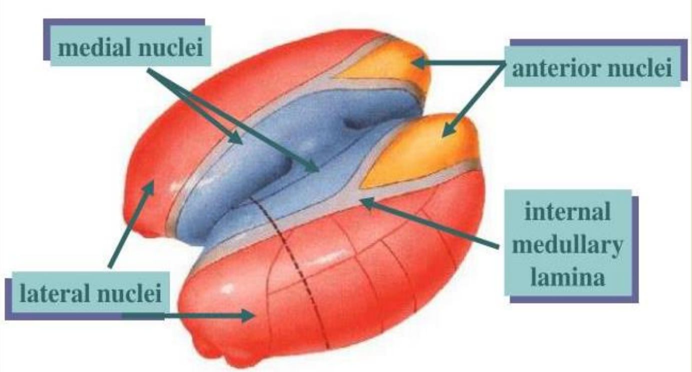

Nucleus basalis of Meynert

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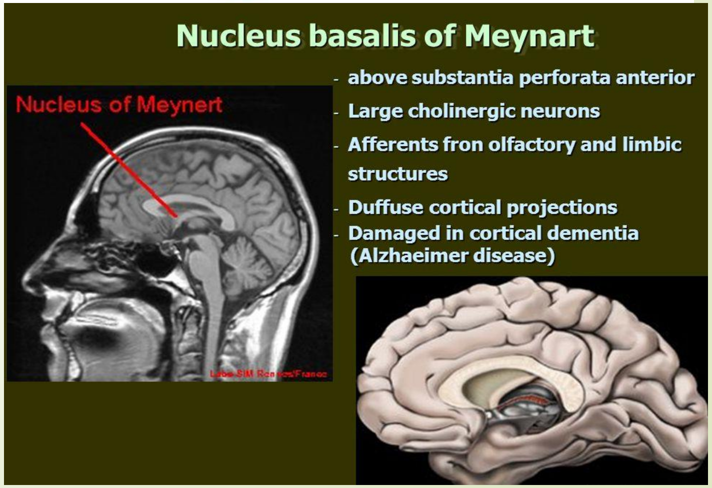


Bilateral damage is needed to have memory loss

Dementia Generally Classified to Five Subgroups:

- Pure memory loss
- Cortical dementias
- Subcortical dementias
- Mixed Dementias
- Reversible / Treatable Dementias

Pure memory loss: causes

- **Herpes encephalitis sequelae** - bilateral medial temporal lobe lesions
- **Korsakoffs amnesia** - bilateral dorsal median nuclei and mammillary body damage due to chronic thiamine deficiency
- **Vertebral - basilar infarcts** - if result in bilateral temporal lobe lesions
- **Transient global amnesia** – TIA / migrain
- **Limbic encephalitis** - damage to hippocampus/limbic pathway

Wernicke–Korsakoff disease

- A rare but important effect of chronic alcohol misuse.
- This organic brain disorder results from damage to the mammillary bodies, dorsomedial nuclei of the thalamus and adjacent areas of periventricular grey matter; it is caused by a deficiency of thiamine (vitamin B1), most commonly from longstanding heavy drinking and an inadequate diet.
- It can also arise from malabsorption or even protracted vomiting.
- Without prompt treatment, the acute presentation of Wernicke's encephalopathy (nystagmus, ophthalmoplegia, ataxia and confusion) can progress to the irreversible deficits of **Korsakoff's syndrome** (severe short-term memory deficits and confabulation, and also reduced red blood cell transketolase).
- It must be considered in any confused patient; if in doubt, treat anyway.
- Prevention of the Wernicke–Korsakoff syndrome requires the immediate use of high doses of thiamine, which is initially given parenterally and then orally.

Two subtypes of Cortical dementias

- **Alzheimer's disease**
- **Frontotemporal Dementia** (Pick's disease)

Fronto-temporal dementia

- Fronto-temporal dementia This term encompasses a number of different syndromes, including **Pick's diseases** and primary progressive aphasia; all are much rarer than Alzheimer's disease.

- Pathologically there is **focal atrophy of the frontal &/or anterior temporal lobes**

- Patients may present with personality change due to frontal lobe involvement with language disturbance due to temporal lobe involvement.
- Memory is relatively preserved in the early stages.

- Clinically **similar to Alzheimer's** in the **late** stages

- There is no specific treatment.

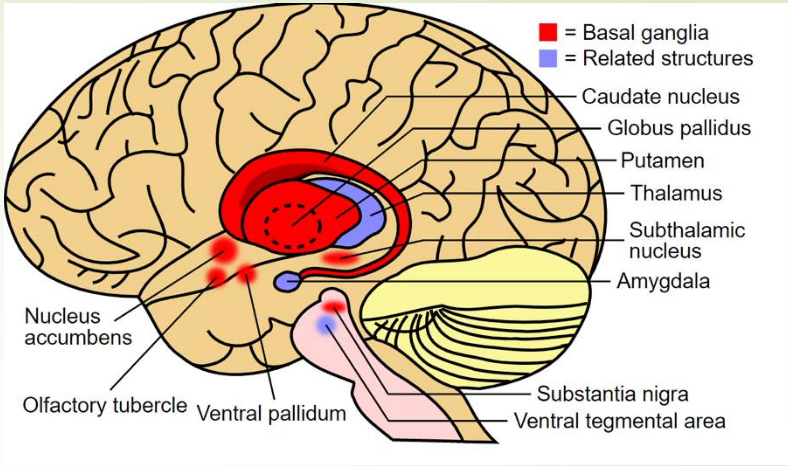
Disorders associated with subcortical dementia:

- Lewy Body dementia
- Parkinson’s disease
- Huntington’s disease
- Progressive supranuclear palsy
- Wilson’s disease
- AIDS dementia

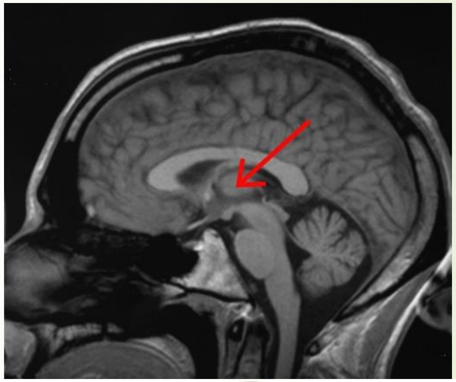
Pathology of subcortical dementia

Subcortical dementia pathologically involves:

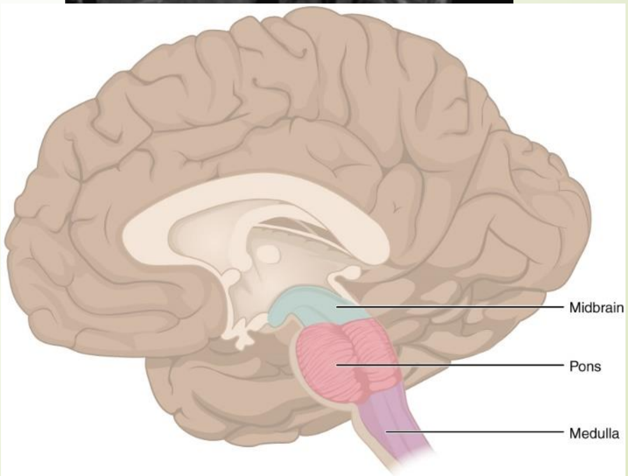
- Basal ganglia =====➔



- Thalamus =====➔



- Rostral brainstem =====➔



Hallmarks of Subcortical dementias

- **Psychomotor retardation**
  - Slowed speech & cognition
  - Motor sequencing problems
  - Abnormal posture and gait
- **Forgetfulness**
  - Characterized by difficulty retrieving information (recall often improves if provide a clue)

Lewy body dementia

- This is a neurodegenerative disorder clinically characterized by dementia and **signs of Parkinson's disease**.
- The cognitive state often fluctuates and there is a high incidence of **visual hallucinations**.
- Affected individuals are particularly sensitive to the **side-effects** of **anti-Parkinsonian** medication and also to **antipsychotic** drugs.
- There is no specific treatment but anticholinesterase (e.g. **donepezil**) agents may well be helpful.

Distinguishing Features of Cortical & Subcortical Dementia

Function	Cortical	Subcortical
Language	Aphasia	Normal
Articulation	Normal	Dysarthric
Praxis	Apraxic	Normal
Agnosis	Present	Absent
Memory	Learning deficits	Retrival deficits
Cognitive tasks	Broken down	Slow, improve with time
Motor	Normal	Abnormal posture & gait

Mixed dementias

Cortical
&
Subcortical structures are both involved

Mixed dementias

1. Multi-infarct dementia (Vascular dementia)

2. Prion Diseases; eg: Creutzfeldt Jacob Disease (CDJ)

3. General Paresis (Neurosyphilis)

Vascular Dementia or Vascular Cognitive Impairment (VCI)

- **VCI** particularly that associated with arteriolosclerotic small vessel disease, has a cognitive profile that is characterized by prominent impairments in executive function and processing speed.

- A **step-wise** progression(**Multi-infarct Dementia)** in association with recognized or unrecognized strokes is characteristic, but a more **gradual progression** is also reported, not infrequently with concomitant AD.

- In many cases, a clinically diagnosed stroke appears to initiate the onset of what has been called **post-stroke dementia**.
- Signs and symptoms related to diagnosed or undiagnosed stroke may also be present.
- **MRI** will show evidence of vascular disease; however, the extent of changes does not clearly correlate with the severity of dementia and its post-stroke progressive course.

Multi-infarct Dementia charactererstics:

- Step-wise progression is hallmark
- focal signs
- By treating stroke risk factors may prevent progression

Creutzfeldt Jacob Disease (CDJ)

CJD is a rare disease, occurring in approximately 2/ million. It is one of the *transmissible spongiform encephalopathies.*

C/F:

- **Prodrome**: fatigue, depression, weight loss, poor appetite ----
- Changes in behavior, emotional response, & intellectual function.
- Ataxia, visual disturbances & dysarthria.
- Most pts present with **rapidly progressive dementia**, followed by muted state, then progress to stupor & coma.
- **Myoclonus** usually develops in > 80% of pts with CJD.
- Spasticity & hyperreflexia are present in > 50% of pts.

Work up:

CSF: ***protein 14-3-3*** is the diagnostic test (sensitivity of 96% & specificity of 99%).

EEG: slow & sharp-wave complexes (1 to 2 Hz) on a slow & low-voltage background.

MRI: nonspecific bilateral ↑ signal on basal ganglia in approximately 80% of pts.

Normal Pressure hydrocephalus

- Clinical triad:

  1) Ataxia, 2) Incontinence, 3) Dementia

- Difficult to identify those who may improve with shunt.
- High morbidity
- Aspiration of Large Volume of CSF is helpful for Dx & Prognosis.(For Benefit from V-P Shunt)

Rapidly progressive dementia

- CJD
- HASHIMATOUS HYPOTHYRODISM
- MULTIINFARCT
- SDH RECURRENT
- INFECTIOUS ENCEPHALOPATHY
- TRAUMA
- HYPOXIC ISCHEMIC ENCEPHALOPATHY

Workup of the demented patient

A. **History** - important to interview family

B. **Physical** and **Neurologic** examination

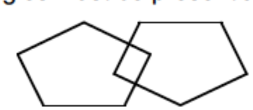
C. **Mental status Examination Tests**

- Mini-Mental Status Examination (**MMSE**)
- Abbreviated Mental Test Score (**AMTS**)

Mini-Mental State Examination (MMSE)

Patient's Name: ______________________________ Date: __________

Instructions: Score one point for each correct response within each question or activity.

Maximum Score	Patient's Score	Questions
5		"What is the year? Season? Date? Day? Month?"
5		"Where are we now? State? County? Town/city? Hospital? Floor?"
3		The examiner names three unrelated objects clearly and slowly, then the instructor asks the patient to name all three of them. The patient's response is used for scoring. The examiner repeats them until patient learns all of them, if possible.
5		"I would like you to count backward from 100 by sevens." (93, 86, 79, 72, 65, …) Alternative: "Spell WORLD backwards." (D-L-R-O-W)
3		"Earlier I told you the names of three things. Can you tell me what those were?"
2		Show the patient two simple objects, such as a wristwatch and a pencil, and ask the patient to name them.
1		"Repeat the phrase: 'No ifs, ands, or buts.'"
3		"Take the paper in your right hand, fold it in half, and put it on the floor." (The examiner gives the patient a piece of blank paper.)
1		"Please read this and do what it says." (Written instruction is "Close your eyes.")
1		"Make up and write a sentence about anything." (This sentence must contain a noun and a verb.)
1		"Please copy this picture." (The examiner gives the patient a blank piece of paper and asks him/her to draw the symbol below. All 10 angles must be present and two must intersect.) 
30		TOTAL

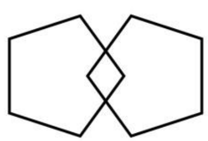
MINI MENTAL STATE EXAMINATION (MMSE)

Name:

DOB:

Hospital Number:

One point for each answer	DATE:		
ORIENTATION Year Season Month Date Time	....../ 5	....../ 5	....../ 5
Country Town District Hospital Ward/Floor	....../ 5	....../ 5	....../ 5
REGISTRATION Examiner names three objects (e.g. apple, table, penny) and asks the patient to repeat (1 point for each correct. THEN the patient learns the 3 names repeating until correct).	....../ 3	....../ 3	....../ 3
ATTENTION AND CALCULATION Subtract 7 from 100, then repeat from result. Continue five times: 100, 93, 86, 79, 65. (Alternative: spell "WORLD" backwards: DLROW).	....../ 5	....../ 5	....../ 5
RECALL Ask for the names of the three objects learned earlier.	....../ 3	....../ 3	....../ 3
LANGUAGE Name two objects (e.g. pen, watch).	....../ 2	....../ 2	....../ 2
Repeat "No ifs, ands, or buts".	....../ 1	....../ 1	....../ 1
Give a three-stage command. Score 1 for each stage. (e.g. "Place index finger of right hand on your nose and then on your left ear").	....../ 3	....../ 3	....../ 3
Ask the patient to read and obey a written command on a piece of paper. The written instruction is: "Close your eyes".	....../ 1	....../ 1	....../ 1
Ask the patient to write a sentence. Score 1 if it is sensible and has a subject and a verb.	....../ 1	....../ 1	....../ 1
COPYING: Ask the patient to copy a pair of intersecting pentagons	....../ 1	....../ 1	....../ 1
TOTAL:	....../ 30	....../ 30	....../ 30



MMSE scoring

24-30: no cognitive impairment
18-23: mild cognitive impairment
0-17: severe cognitive impairment

OME Oxford Medical Education

- The maximum MMSE score is 30 points.
- On average Alzheimer's declines about two to four points each year.
- MMSE can help in staging of Alzheimer's dis **severity** as follow:

0-9	Severe dementia
10-20	Moderate dementia
21-24	Mild dementia
25-30	May be normal

Investigations for Dementia:

Lab studies:

- Electrolytes,
- Urea,
- Creatinine,
- Ca++,
- CBC,
- Thyroid studies,
- ESR,
- B 12,
- Folate
- VDRL
- HIV Ab, &
- Drug screen if appropriate

Other studies:

- ECG,
- CXR,
- Brain CT,
- MRI,
- SPECT,
- PET Scan, &
- LP : for infectious etiology

Take Home Massage

- Dementia is an acquired chronic syndrome of intellectual impairment that interferes with social & occupational function.
- Delirium is characterized by a disturbance of consciousness and a change in cognition that develop over a short period of time.
- Cortical & subcortical dementia can be differentiated clinically to a great extent.
- Many cause may operate together in mixed dementia.
- Mental status Examination Tests help in Diagnosis & staging & follow up.
- Workup to identify & treat reversible cause is necessary.

THE FOLLOWING ARE TRUE EXCEPT

a) AD IS COMMON TYPE OF DEMENTIA
b) WILSON DISEASE IS NOT CURABLE TYPE
c) CHRONIC ALCOHOLISM MAY END WITH DEMENTIA
d) NO AVAILABE TREATMENT FOR MOST TYPES OF DEMENTIA